

Yuulin An

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Department of Economics, Colorado State University
1771 Campus Delivery, Fort Collins, CO 80523

RESEARCH INTERESTS & SKILLS

Fields Health, Environmental, Demographic Economics

Software R, Python, Stata, Git/GitHub, $\text{\LaTeX}$

Causal Inference Difference-in-Differences, Event Studies, Regression Discontinuity, Instrumental Variables

Language Khmer (native), English (fluent)

EDUCATION

Colorado State University

Ph.D., Economics (Candidate)

M.A., Economics

Fort Collins, CO

Expected Spring 2027

May 2024

Coe College

B.A., Economics and Business Administration (Cum Laude)

Cedar Rapids, IA

May 2022

JOB MARKET PAPER

Airborne Lead and Birth Outcomes: Evidence from Leaded Fuel in Auto Racing

Abstract: The National Association for Stock Car Auto Racing (NASCAR) and Automobile Racing Club of America (ARCA) races continued to use leaded gasoline until 2007, despite the removal of leaded gasoline from on-road vehicles long ago. Exploiting this novel quasi experiment, I study how airborne lead exposure affected birth outcomes using administrative birth records linked to race events at Texas Motor Speedway (TMS) in Fort Worth, Texas. Mothers living within 25 km of TMS see a reduction in infant birthweight, shortened gestation length, and an increased probability of low birthweight before the ban of lead from NASCAR and ARCA races, while a post-ban placebo yields coefficients near zero. To translate race-day exposures into blood lead burdens, I developed a pregnancy-specific physiologically based pharmacokinetic (PBPK) model that simulates maternal and fetal daily blood lead trajectories. I find that a one-standard-deviation increase in second-trimester PBPK blood lead reduces birthweight by approximately 73 grams. A potential mechanism suggests that lead increases gestational hypertension and shortens gestation, and these effects disappear after the transition to unleaded fuel. These findings suggest that even a relatively low level of lead can have a meaningful effect on maternal and infant health.

PEER-REVIEWED ARTICLES

An, Y., Chin, S. & Miller, R. (2025). Growing Old in Rural America: Measuring Late Life Health and Economic Well-being. *The Journal of the Economics of Ageing*, 31, 100565.

WORKING PAPERS

Population Aging, Local Economic Growth, and the Urban-Rural Divide, with Tim Komarek and Ray Miller. Under Review at *Regional Science and Urban Economics*.

Maternal Lead Exposure and Birth Outcomes: Evidence from Natural Experiments in the Phase-out of Leaded Gasoline, with Sammy Zahran, Christopher Keyes, Sayorn Chin, and Susan Averett.

SELECTED WORKS IN PROGRESS

When Football Comes to Town: Provider Decision-Making and Delivery Complications, with Sayorn Chin and Sammy Zahran.

Urban Heat Island and Health, with Tim Komarek, Sammy Zahran, and Sayorn Chin.

OTHER PUBLICATIONS

Population Aging and Economic Growth in Colorado: Why Regional Connectivity Matters for Rural Communities, with Ray Miller and Tim Komarek. *Regional Economic Development Institute*, March 2026.

TEACHING EXPERIENCE

Department of Economics, Colorado State University

Fort Collins, CO

Graduate Teaching Instructor

Aug. 2025 - present

- Summer 2026: Working with Data (1 section)
- Spring 2026: Working with Data (2 sections)
- Fall 2025: Working with Data (2 sections)
- Summer 2025: Working with Data (1 section)

Department of Economics, Colorado State University

Fort Collins, CO

Graduate Teaching Assistant

Aug. 2022 - 2025

- Spring 2025: Principles of Macroeconomics (3 sections)
- Fall 2024: Principles of Microeconomics (3 sections)
- Spring 2024: Principles of Microeconomics (3 sections)
- Fall 2023: Principles of Macroeconomics (3 sections)
- Spring 2023: Principles of Microeconomics (3 sections)
- Fall 2022: Principles of Microeconomics (3 sections)

RESEARCH EXPERIENCE

Regional Economic Development Institute

Fort Collins, CO

Research Assistant to Dr. Ray Miller

Summer 2025

Future of Work, Russel Sage Foundation

Fort Collins, CO

Research Assistant to Dr. Zachary Schaller

Summer 2024

HONORS & AWARDS

Dissertation Fellowship

Fall 2026

Awarded a departmental Dissertation Fellowship to support accelerated progress on dissertation research in the Fall semester.

WEAI Graduate Student Workshop

Summer 2026

Selected for the highly competitive Western Economic Association International's Graduate Student Workshop, focused on job market paper presentation and academic career preparation.

NBER Mentoring Program on Aging and Health Economics Research

May 2024

Selected for NBER's mentoring program in aging and health economics supported by the National Institute on Aging.

Graduate Teaching Assistantship

2022 – 2027

Awarded a full tuition scholarship and monthly stipend from the Department of Economics at Colorado State University.

Harley W. Howell Prize

May 2022

This award is given to a senior business and economics major for superior academic achievement, promise of leadership career in the field, sound moral character and campus citizenship (issued by the Stead Department of Business Administration & Economics from Coe College).

Spellman Optimum Optimorum

May 2022

This award is given to a senior who will be entering an economics Ph.D. program immediately following their graduation from Coe College.

CONFERENCES

Western Economic Association International, <i>Participant/Presenter</i>	Summer 2026
North American Regional Science Council (NARSC), <i>Presenter</i>	Fall 2025
NBER Summer Institute Economics of Health Conference, <i>Participant</i>	Summer 2025
NBER Racial and Ethnic Health Disparities Conference, <i>Participant</i>	Spring 2024

SERVICES

Brown Bag Seminar Series	Aug. 2024 - May 2025
Organizer	

PROFESSIONAL MEMBERSHIPS

Columbine Health Systems Center for Healthy Aging at Colorado State University
Western Economic Association International
American Economic Association
American Society of Health Economists

REFERENCES

Ray Miller Associate Professor Department of Economics Colorado State University ray.miller@colostate.edu	Sammy Zahran Professor Department of Economics, Colorado State University Department of Epidemiology, Colorado School of Public Health sammy.zahran@colostate.edu
Jackie Young Research Associate II National Wildlife Research Center (NWRC) Colorado State University jackie.young@colostate.edu	